

An Early Warning System Based on Visual Feedback for Light-Based Hand Tracking Failures in VR Head-Mounted Displays

Mohammad Raihanul Bashar , and Anil Ufuk Batmaz 

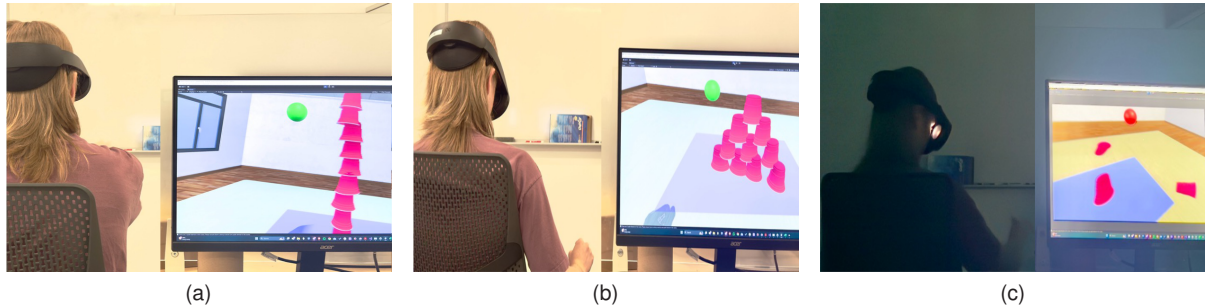


Fig. 1: A participant is performing a pick-and-place task (Cup Stacking) within a VR environment using their bare hand. a) The image shows that the participant completed the task by successfully stacking the cups in a pillar shape. b) The participant completed the task by successfully stacking the cups in a triangular shape. c) The hand tracking failure scenario is introduced by dimming the light completely (the screen light was dimmed throughout the experiment, with additional lighting added artificially for illustration purposes). The system warned the user by changing the color of the sphere in front of them *before* the built-in hand tracking failed.

Abstract—State-of-the-art Virtual Reality (VR) Head-Mounted Displays (HMDs) enable users to interact with virtual objects using their hands via built-in camera systems. However, the accuracy of the hand movement detection algorithm is often affected by limitations in both camera hardware and software, including image processing & machine learning algorithms used for hand skeleton detection. In this work, we investigated a visual feedback mechanism to create an early warning system that detects hand skeleton recognition failures in VR HMDs and warns users in advance. We conducted two user studies to evaluate the system's effectiveness. The first study involved a cup stacking task, where participants stacked virtual cups. In the second study, participants performed a ball sorting task, picking and placing colored balls into corresponding baskets. During both of the studies, we monitored the built-in hand tracking confidence of the VR HMD system and provided visual feedback to the user to warn them when the tracking confidence is 'low'. The results showed that warning users before the hand tracking algorithm fails improved the system's usability while reducing frustration. The impact of our results extends beyond VR HMDs, any system that uses hand tracking, such as robotics, can benefit from this approach.

Index Terms—Hand Tracking, Virtual Reality, Visual Feedback, System Usability, Early warning

1 INTRODUCTION

Virtual Reality (VR) Head-Mounted Displays (HMDs) immerse users in virtual environments, offering (close-to) real-time interactive experiences. To achieve this, VR HMDs use a combination of sensors and devices to capture and interpret user movements in the real world and map these interactions in the virtual environment. The effectiveness of VR HMDs tracking performance depends on various tracking hardware, such as controllers, trackers, and integrated cameras, and software, such as image processing, machine learning, and filtering algorithms. Together, these components work to accurately monitor and translate user movements and interactions from the real world to the virtual environment. Among the various tracked movements, the user's hands are one of the most frequently monitored, as precise hand tracking is crucial for intuitive interaction within VR environments.

Interacting with the virtual environment using one's hands feels

highly engaging, while it also fosters a greater sense of ownership compared to using hand held controllers [44]. With "inside-out" or "built-in" tracking via RGB or infrared cameras in current consumer models, HMDs can track hand movements without the need for external devices like wearable trackers, when the hands are positioned roughly in front of the headset. This reduces costs and increases portability and user-friendliness, making VR applications more accessible. However, the effectiveness of hand tracking is highly dependent on hardware, software, and environment [72].

The hand tracking systems that use cameras or optical sensors face several challenges due to, e.g., different light sources [16], camera resolution [4], levels of brightness [30], and objects blocking the view [16]. These challenges can significantly affect how well hand tracking algorithms perform. Many existing commercial systems now use a combination of multiple tracking systems, and this approach has demonstrated increasing algorithm robustness [58]. However, even systems that use a combination of tracking technologies are not completely reliable in practical application settings when users interact with virtual objects with their bare hands [16, 45].

In practical application scenarios, when a tracking system encounters challenges and loses hand tracking, the common approach is to leave the user's avatar's hand in its last tracked position or completely stop rendering the hands. When tracking resumes, the virtual hand abruptly jumps to the new position based on the user's recent hand position, creating a noticeable discontinuity in motion. This prominent error

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can reduce the perception of realism, disrupt task execution, and negatively affect the user experience [65]. Furthermore, these failure events could decrease the user's trust in the system, elevate frustration levels, and subsequently increase the error rate and reduce the precision [65]. Additionally, manual corrections in tasks requiring precision, like positioning a medical screw during surgical planning in VR [52], can be time-consuming, yielding increased task execution times. In such scenarios, users may need to restart the task if the tracking algorithm encounters failures.

In recent studies, hand tracking error correction mechanisms have been explored to improve user experience and hand tracking reliability in virtual reality environments. For example, Abdulkarim et al. [1] proposed a lens distortion based error-correction method, and Huang et al. [34] focused on improving the finger movement tracking by using a Hidden Markov Model. These studies focus on providing information to users following the occurrence of tracking errors, aiming to mitigate the impact of such errors on task performance and system usability. Yet, these studies propose solutions *after* the hand tracking algorithms fails.

In this paper, using the built-in hand tracking quality data, we aim to introduce and evaluate an early warning system based on visual feedback mechanisms to mitigate the negative effects of hand tracking algorithm failures within virtual reality systems *before* the hand tracking system failure occurs, i.e., providing in advance warnings to users enables them to prepare for system failures mentally. The proposed method attempts to enhance the usability of immersive VR HMDs and improve the overall user experience. In summary, the contributions of this research are:

- The development of an early warning system that provides users with visual feedback regarding hand tracking failures when interacting within virtual environments using their bare hands.
- Showing that the implementation of this early warning system significantly improves system usability and reduces user frustration levels, as demonstrated by results from the System Usability Scale (SUS) and NASA Task Load Index (TLX) questionnaires.

2 PREVIOUS WORK

2.1 Hand Tracking

Previous work has proposed various interaction techniques that rely on input devices and user gestures, each with their own taxonomies and evaluation methodologies [5, 12–14, 26, 36, 49]. A straightforward and natural method of interaction within virtual reality systems is the use of bare hands and mapping the movements of the real world hand to the virtual hands [40, 55]. Systems, such as the Meta Quest Pro/3, HTC Vive Pro 2, and Pico Neo 4, offer users the option between hand tracking and controller tracking. Others, like Microsoft HoloLens and Apple Vision Pro, rely exclusively on hand tracking. The appeal of using bare hands instead of handheld controllers lies in the potential for a more natural interaction interface. With recent advancements in computer vision, integrating visual hand tracking has become a standard feature in VR HMDs, allowing for flexible and expressive spatial interactions [41].

The previous work introduced new hand tracking systems and algorithms that can be used in virtual environments. For instance, Dente et al. [22] presented a tracking method based on complex wavelet orientation fields to detect hand and finger movements. Toby et al. [63] proposed a hand tracking system using a single depth camera. Shamaie et al. [62] introduced an algorithm for tracking bimanual movements. Prisacariu and Reid [57] developed a 3D hand tracking method that combines a region and model-based tracker with an accelerometer attached to the hand, demonstrating robustness to motion blur, occlusion, and cluttered environments, though their method struggles with recognizing the correct hand pose. Weng et al. [69] introduced a real-time hand tracking and posture recognition algorithm using a multi-cue approach that is both scale-invariant and rotation-invariant. Ghosh et al. [28] proposed a vision-based markerless scale-invariant hand tracking method capable of tracking multiple hands with a greater degree of freedom compared to previous methods.

User engagement and presence in virtual world increase with the capability of physical interactions [37]. Pei et al. [54] demonstrated that using hand tracking for interaction is advantageous as it is perceived as more natural and intuitive. Additionally, virtual hand has been shown to contribute positively to features such as enjoyment, and presence [2]. The accurate tracking of hands not only enhances user immersion [6] but also allows for more precise and accurate movements, particularly significant in applications like training scenarios [31]. Since the virtual hand is one of the crucial techniques for natural and intuitive interactions in VR [43], we investigated user performance and user experience with our approach using the virtual hand interaction technique.

2.2 Hand Tracking Failures

Across various fields and applications, hand tracking encounters numerous challenges that can lead to a potential loss of tracking precision. Despite advances in technology and theoretical approaches, achieving robust, accurate, and real-time hand tracking via vision based marker-less tracking system is a challenging problem [3]. Lighting conditions, occlusions, depth ambiguities, background clutter, noise in the captured image, fast and non-linear hand motion, insufficient distinguishing visual cues between different parts of the hand, and the computational complexity of real-time processing are some of the major challenges of marker-less hand tracking systems. These challenges become even more complex when hands interact with each other or with objects in the virtual environment, even when multiple cameras are used to track the hand movement. Voigt-Antons et al. [66] compared hand tracking and controllers and found that hand interaction exhibited lower precision than controller interaction due to inherent challenges in hand tracking performance. Huang et al. [35] also reported that user's performance was similar between controller and hybrid interaction (bare hand and controller combined) compared to just bare hand interaction. Further studies [40, 48] supported these findings, emphasizing that the precision of hand tracking is often compromised by factors such as sensor resolution and the robustness of the tracking algorithms.

These challenges significantly affect the overall perception and performance of users when using VR systems. Poor hand tracking precision can disrupt the sense of immersion, as users become aware of the system's limitations, which take away from the seamless and natural interaction that VR aims to provide [65]. Moreover, the lack of reliable hand tracking can hinder the completion of precise tasks, thereby limiting the range of applications for which VR can be effectively utilized. In this paper, we investigated the effects of hand tracking failures on users' perceptions of virtual reality environments.

2.3 Visual Feedback

The human brain integrates sensory data to form a coherent understanding of the environment, influencing motor responses. In VR, controlled sensory inputs can alter perceptual information, affecting task performance that requires motor control. Without the tactile feedback of a controller, interacting with virtual objects using one's hands can lead to slower interactions, as users may not clearly perceive when contact or manipulation has occurred. Providing feedback can make this interaction faster and more accurate. For instance, Mestre et al. [50] explored the effect of feedback within VR system for a virtual reality exercise system. Douglas et al. [23] investigated the effect of visual feedback on user performance for a color picker interface and found that it improves user performance, similar to the findings of Burke et al. [17]. Visual feedback plays a crucial role in enhancing user performance and perception in both real and virtual environments. This is particularly true in VR, where it directly influences how users perceive and interact with the virtual world. Bowman et al. [15] highlighted that inadequate visual feedback in VR systems can disrupt user engagement and lead to higher error rates.

Studies have shown that effective visual feedback can significantly improve task performance by providing real-time information about errors and corrections needed [20, 61]. Prior research also used visual error feedback to inform the user if they needed to improve their performance [42, 73]. For example, Wang et al. [68] demonstrated that visual collision feedback improved hand movement accuracy when interacting

in an occluded virtual reality environment. Similarly, Cooper et al. [20] found that real-time visual feedback helps users correct their actions more efficiently, thereby enhancing overall performance.

Furthermore, Kosch et al. [42] compared different types of visual feedback and found that users who received real-time error notifications performed significantly better than those who did not. Zhang et al. [73] also emphasized the importance of visual error feedback in VR training systems, where immediate visual cues helped users to adapt and refine their motor responses quickly. Our work explores the integration of visual feedback as an early warning mechanism to inform users about hand tracking failures during bare hand interactions in VR systems.

3 MOTIVATION AND HYPOTHESIS

Previous work showed that users find tasks easier to complete when they can observe their virtual hands moving and interacting in sync with their real-world actions, maintaining realistic proportions and details [48]. However, if the hand tracking algorithm of the headset fails to track the hand due to factors such as low light intensity, hand movements can lead to unnatural or glitchy hand movements in the virtual environment or even not rendered. This breaks immersion and reduces user confidence in the VR setting [29]. If users can be warned before the tracking failure occurs, they can be mentally prepared for failure, and it can improve user experience. To the best of our knowledge, no such visual feedback based warning system exists for hand tracking failures. In this paper, our goal is to investigate the effectiveness and usability of an early warning system for hand gesture recognition/tracking failure. In this work, we hypothesize that:

H1. An early warning system with visual feedback improves the usability of the virtual reality system: Since users have to manipulate and interact with objects in the VR with their hands, a warning system can allow them to get mentally prepared for the failure in advance. Before/during the failures, users can adjust their poses or movements accordingly, which can increase system usability. To assess this, we measured usability via the System Usability Scale (SUS) [39], as used in previous studies [53,66].

We also hypothesize that **H2. An early warning system reduces the user's perceived task workload while interacting using bare hand within a virtual environment:** Previous research has investigated the impact of warning systems on users' subjective mental workload, finding that such systems reduce mental workload and enhance users' overall system awareness [67]. Applying the early warning system before the hand tracking system fails in VR HMDs can also decrease the users' perceived task workload. To evaluate this, we analyzed the NASA Task Load Index (TLX) [32] data to quantify subjective workload across our experimental conditions.

4 USER STUDIES

In order to evaluate the effectiveness of the proposed methods, we conducted two user studies, each using a different task. All ethical and experimental procedures were approved by the Concordia University Human Research Ethics Board (Ethics Protocol Number: 30019459).

- **Study 1 - Cup Stacking:** Study 1 focuses on a pick-and-place task where participants had to stack virtual cups in two different pre-defined shapes (Fig. 1). The stacking task has been selected to simulate real-world activities that require careful manipulation and spatial arrangement of objects [21,59], making it an ideal candidate for testing the usability and reliability of an early warning system for hand tracking failures.
- **Study 2 - Ball Sorting:** Study 2 focuses on another pick-and-place task where participants were required to pick up balls and place them in the corresponding colored baskets, see Fig. 5. This task simulates real-world scenarios where users must quickly and accurately sort objects based on visual characteristics, a common requirement in various VR applications, such as training and simulation environments [38]. The ball sorting task allows for the assessment of early warning system in dynamic, goal-directed

tasks, evaluating effectiveness of system in reducing perceived task workload and improving overall user experience.

4.1 Study 1 - Cup Stacking

4.1.1 Participants

For a within-subjects study, we recruited 18 participants (fifteen male, three female) with ages ranging between 22 and 39 ($M = 24.17$, $SD = 6.72$). All participants either had normal vision or vision corrected to normal. Participants came from diverse backgrounds, including engineering, advertising, and customer service, offering a broad range of perspectives for the evaluation. None of them reported having color vision deficiencies or any other visual impairments. Out of the 18 participants, eleven mentioned the right eye, and seven mentioned the left eye as their dominant eye, and all of them were right-handed. When they were asked about their VR experience, seven participants mentioned none, five reported once, two reported 3 times, and four reported experiencing VR more than 5 times in their lifetime.

4.1.2 Apparatus

The experiment was performed on a desktop PC with a 13th Gen Intel(R) Core(TM) i9-13900KF processor running at 3.0 GHz, 32 GB RAM, and an NVIDIA GeForce RTX 4070 graphics card. We chose Meta Quest Pro as VR head-mounted display. We created the virtual scene in Unity version 2021.3.24 and the Oculus Unity-Integration SDK v57.0.1-deprecated. The Unity application used in the study was executed via Meta Quest Link. We used the VR HMDs built-in hand tracking mechanism instead of external hardware such as gloves.

4.1.3 Procedure

We used ten virtual cups for virtual objects and asked participants to build a pillar and a triangular shape S_2 : **pillar and triangular shape** by stacking these cups. The study involved three different tracking failure conditions: TF_3 : **no failing, with feedback, and without feedback**.

Upon arrival, participants were greeted and briefed on the task, which involved stacking cups in a virtual reality environment using virtual hand interaction. Then, they signed a consent form and completed a questionnaire to provide their demographic information. Afterward, the participants were seated comfortably in a chair facing a blank wall and equipped with the Meta Quest Pro VR headset Fig. 1. The placement in front of a blank wall was intentional to ensure that no environmental factors, such as screen monitor lights, would interfere with the hand tracking, particularly during instances when the lights were turned off to introduce tracking failures. The experiment room had no windows or other sources of light.

Before starting the experiment, participants were given time to familiarize themselves with the virtual environment. They practiced interacting using their bare hand within the virtual environment until they felt confident and ready to proceed with the actual experiment. The user study lasted approximately an hour, including the time participants spent familiarizing themselves with the virtual environment and hand tracking.

4.1.4 Virtual Environment

In the virtual environment, participants were placed inside a virtual room with monocular visual cues, e.g., texture gradient, linear lines, and depth cues to simulate depth and spatial relationships. Texture gradients appeared on surfaces like the wooden table and the walls, where details become denser and less distinct in the distance, creating an impression of depth. Linear perspective was reinforced by parallel edges on the table or floor tiles converging toward a vanishing point, heightening the perception of spatial dimensionality. The room contained a wooden table directly in front of the participants, with ten upside-down cups arranged. A blue rectangular area indicated the designated spot for the stacking task. In addition, two virtual spheres (each with a radius of $0.5cm$) were positioned $35cm$ away from the user's eye, $0.7cm$ above the user's eye level—one on the left and one on the right. These spheres served as indicators of each hand's tracking status. The most reliable hand tracking occurs when hands are within the field of view of the VR headset's inside-out cameras, which typically cover the area in

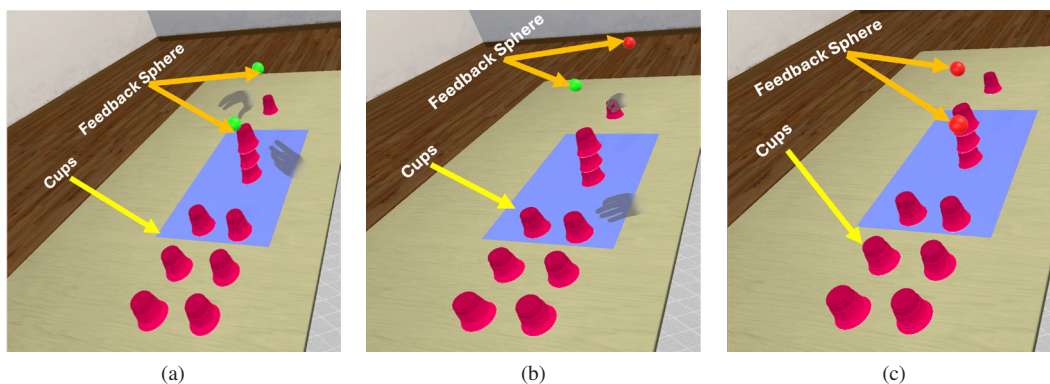


Fig. 2: a) The hand tracking algorithm can properly track both hands and the feedback spheres are green (highlighted in orange arrow, cup highlighted in yellow). b) The built-in hand tracking algorithm returns a *low* confidence value for the right hand and a *high* one for the left hand, so the feedback sphere for the right is red and green for the left. c) The hand tracking algorithm cannot track hands, the feedback spheres are red for both hands, and the system can not render any of the virtual hands, as it can not determine their position and rotation.

front of the user's face [16]. Positioning the feedback spheres slightly above eye level ensured that they were always visible during the task execution but did not obscure the view of the tasks and hands that were below the eye level, regardless of participants' viewing angle or position within the virtual space. Since users tend to focus on their hands during goal-directed tasks, such as the stacking task in this scenario, this setup allowed them to monitor and adjust their movements based on the system's tracking accuracy, leading to a more intuitive interaction. The spheres changed from green to red to reflect the hand-tracking confidence level: green indicated *high confidence* ("go"), while red signaled *low confidence* or failure ("stop"), mirroring a traffic-light color scheme that is widely recognized (see Fig. 2). The built-in HMD SDK provided this confidence level variable, i.e., we did not use additional algorithms to detect hands.

The task involved stacking cups into both pillar and triangular shapes, as seen in Fig. 1, which is a pick-and-place task. This pick-and-place task was specifically chosen due to its simplicity and familiarity, ensuring that it was accessible and easy to understand for all participants. Stacking cups (also known as sports stacking [21]) is a straightforward exercise that requires basic hand-eye coordination and dexterity, skills that most individuals possess from everyday experiences [59]. Previous research has explored cup stacking to analyze the motor performance of individuals in the upper extremities [21, 47, 59, 71]. The straightforward nature of the cup stacking task allows participants to concentrate on the main objectives of the experiment, i.e., interpreting the early warning system feedback, without being distracted by complex instructions or unfamiliar actions. The simplicity of the task also enabled participants to engage with the experiment quickly and provided more reliable and consistent data on their interactions within the virtual environment. Participants used their bare hands without any haptic feedback to interact with the objects in the virtual environment. Each participant performed each task once.

4.1.5 Hand Tracking Failures

In the experiment, hand tracking failures were deliberately and randomly introduced *three times* during both the "with feedback" and "without feedback" conditions. Thus, we did not record an additional error variable. The randomness in the experiment was controlled using a Python script executed concurrently with the Unity application. At three random intervals, the Python script displayed an alert prompting the experimenter to immediately turn off the room lights. After the first alert, the script issued a second alert two seconds later, indicating it is time to turn the lights back on. Because the light switch was within the experimenter's reach, the experimenter could respond quickly to both alerts. Since the lights were turned off manually, we could not measure the exact values, but the experimenter recorded the time for events with a chronometer. Each failure event involved turning the lights off

and then back on in the experiment room. On average, the overall process—receiving the first alert, turning the lights off, receiving the second alert, and then turning the lights back on—took *two seconds* on average.

We used the headset's built-in function to monitor the hand tracking algorithm's performance, which was categorized as either 'low' or 'high.' When the experiment room lights were dimmed, the hand tracking performance decreased from 'high' to 'low.' Despite this change, the algorithm continued to track and render the virtual hands, so participants were warned about the potential for tracking failure. When the experimenter fully dimmed the lights (lights were completely turned off), the hand tracking algorithm failed completely, i.e., the built-in hand tracking system could no longer track or render the participant's virtual hands. During a pilot test, we observed that dimming the lights consistently produced a 'low' performance level but did not cause a total loss of tracking; a complete loss of tracking occurred only when the lights were completely turned off. We also noticed that turning the lights off directly triggered a 'low' response almost immediately, making the latency negligible. Therefore, we used this 'low' indicator to present the warning to participants about the impending hand-tracking failure. We also found that a minimum two-second interval of completely dimmed light (darkness) did not affect the overall task performance. Throughout the experiment, participants wore an HMD that blocked their view of the real environment, meaning they were unable to see the changes in the light conditions and were unaware of when the tracking would fail because of the randomness of the tracking failure process, they can not predict as well.

We counterbalanced the tracking failure condition (TF_3) and stacking shapes (S_2), using a Latin Square design across 18 participants. For each condition, we recorded the time to complete the task. At the end of each condition, we asked participants to fill the System Usability Scale (SUS) [39] and NASA Task Load Index (TLX) [32] questionnaires. In SUS, we asked participants 10 questions, which are provided in the supplementary materials.

4.1.6 Results

We conducted a two-way ANOVA using SPSS 29, with $\alpha = 0.05$, to compare the effects of tracking failure conditions and stacking types. We considered data normally distributed if Skewness (S) and Kurtosis (K) were within ± 1.0 [8, 27]. If the data was not normally distributed even after log transformation, we used Aligned Rank Transform [70] for data analysis. For post-hoc analyses, we used the Bonferroni method.

We used the Friedman Test to analyze SUS and NASA-TLX results and the Wilcoxon Signed-Rank test for post-hoc analyses. The results are presented as means along with the standard error of the means. To maintain brevity, we focus only on the statistically significant results.

Time:

Table 1: Time taken to complete stacking the cups under each tracking failure condition.

Parameters	No Failing		With Feedback		Without Feedback	
Stacking	Pillar Shape	Triangular Shape	Pillar Shape	Triangular Shape	Pillar Shape	Triangular Shape
Time	104.5 ± 3.97	28.61 ± 1.42	104.6 ± 3.62	30 ± 1.34	101.8 ± 2.78	28.4 ± 0.66

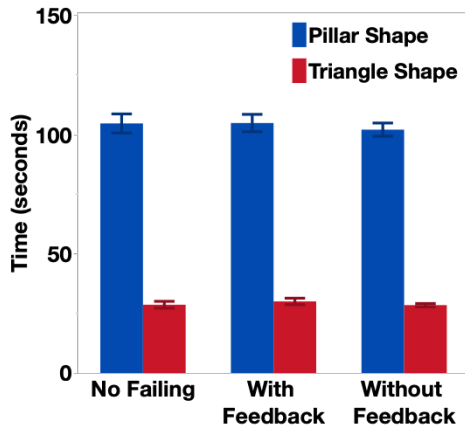


Fig. 3: Task (Cup Stacking) completion time (in seconds) for different tracking failure conditions (No Failing, With Feedback, Without Feedback) and stacking types (Pillar Shape, Triangle Shape).

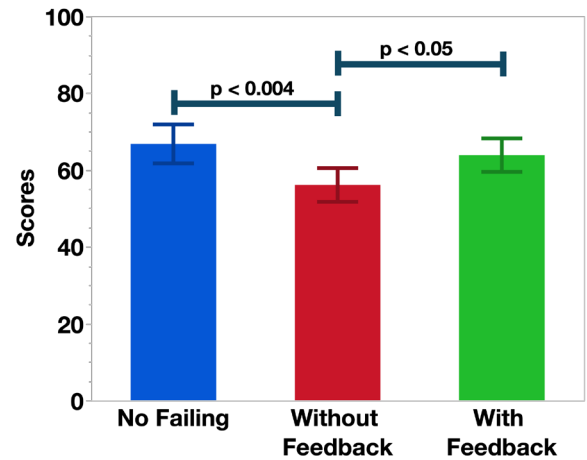


Fig. 4: SUS questionnaire scores for (Study - 1) tracking failures condition: No Failing, With Feedback, and Without Feedback. The questions are provided in supplementary materials.

According to the results illustrated in Tab. 1 and Fig. 3, we did not observe any significant difference between NO FAILING, WITH FEEDBACK AND WITHOUT FEEDBACK in terms of time (measured in seconds). However, there was a significant effect of stacking type, ($F(1, 102) = 308.562, p < .001, \eta^2 = .752$), suggesting that the type of stacking significantly affected task performance. The interaction effect between the feedback type and the stacking type was not significant. According to the results, participants stacked cups faster in triangular shape than in pillar shape.

SUS: Based on the SUS scores presented in Tab. 2, there was a decrease in the SUS score for the WITHOUT FEEDBACK tracking failures condition. We observed an average SUS score of 55 (Grade = D) for WITHOUT FEEDBACK, which is defined as a ‘Poor’ rating. We analyzed the

FEEDBACK condition compared to NO FAILING ($Z = -2.903, p < 0.005, \eta = 0.684$), as illustrated in Fig. 6.

Participants reported higher frustration in the WITHOUT FEEDBACK tracking failures condition compared to the WITH FEEDBACK and NO FAILING tracking failures condition. This suggests that the absence of visual feedback, combined with hand tracking failures, contributed to increased frustration among the participants. These findings align with our hypothesis **H2**, namely that an early warning system reduces the user’s perceived task workload when interacting with a virtual environment using bare hands input.

4.2 Study 2 - Ball Sorting

In the second within-subject user study, we implemented a sorting task where participants placed colored balls into corresponding baskets, as seen in Fig. 5. We chose this task because it has been previously used to evaluate users’ eye-hand coordination performance [7, 11, 60]. The sorting task requires users to separate and categorize objects, adding additional cognitive workload compared to Study 1. Yet, similar to Study 1, the simplicity of this task allowed participants to focus on the main objectives of the experiment, such as perceiving the visual feedback system and executing hand tracking movements, without being distracted by complex instructions. This straightforward approach minimized learning effects and enabled participants to engage quickly, resulting in more reliable and consistent data regarding their interactions within the virtual environment.

4.2.1 Participants

We recruited eighteen new participants (10 male and 8 female) from the local university. Their age ranged between 18 and 37 ($M = 25.27, SD = 3.61$). Similar to Study 1, participants in this study represented a broad range of perspectives. All participants either had normal vision or vision corrected to normal. None of them reported having color vision deficiencies or any other visual impairments. Of the 18 participants, sixteen mentioned the right eye, and two mentioned the left as their dominant eye. All participants were right-handed except for one. When they were asked about their VR experience, six participants mentioned none, two reported once, two reported five times, and eight reported experiencing VR more than 10 times in their lifetime.

Table 2: SUS Scores and Grades for three tracking failures condition.

Condition	Score	Grade
No Failing	67.05	C
With feedback	65.5	C
Without feedback	55	D

overall SUS scores using a Friedman Test, which revealed a significant difference across the three conditions—NO FAILING, WITHOUT FEEDBACK, and WITH FEEDBACK ($\chi^2(2, 18) = 8.149, p < 0.017$). Post-hoc Wilcoxon Signed-Rank Tests revealed a significant difference between NO FAILING and WITHOUT FEEDBACK ($Z = -2.900, p < 0.004, \eta = 0.68$), as well as between WITHOUT FEEDBACK and WITH FEEDBACK ($Z = -1.949, p < 0.05, \eta = 0.46$). However, there was no significant difference between NO FAILING and WITH FEEDBACK ($Z = -0.687, p = 0.492, \eta = 0.16$), as illustrated in Fig. 4. These findings align with our hypothesis, **H1**, that an early warning system enhances VR usability when hand tracking failures occur during bare hands interaction.

NASA TLX: The Friedman Test revealed a significant overall difference in NASA TLX scores across the three conditions—NO FAILING, WITHOUT FEEDBACK, and WITH FEEDBACK ($\chi^2(20, 18) = 55.431, p < 0.001$). Further analysis of the individual workload subscales showed that frustration levels differed significantly among the conditions ($\chi^2(2, 18) = 7.115, p < 0.029$). The Wilcoxon Signed-Rank Test indicated that frustration was significantly higher in the WITHOUT

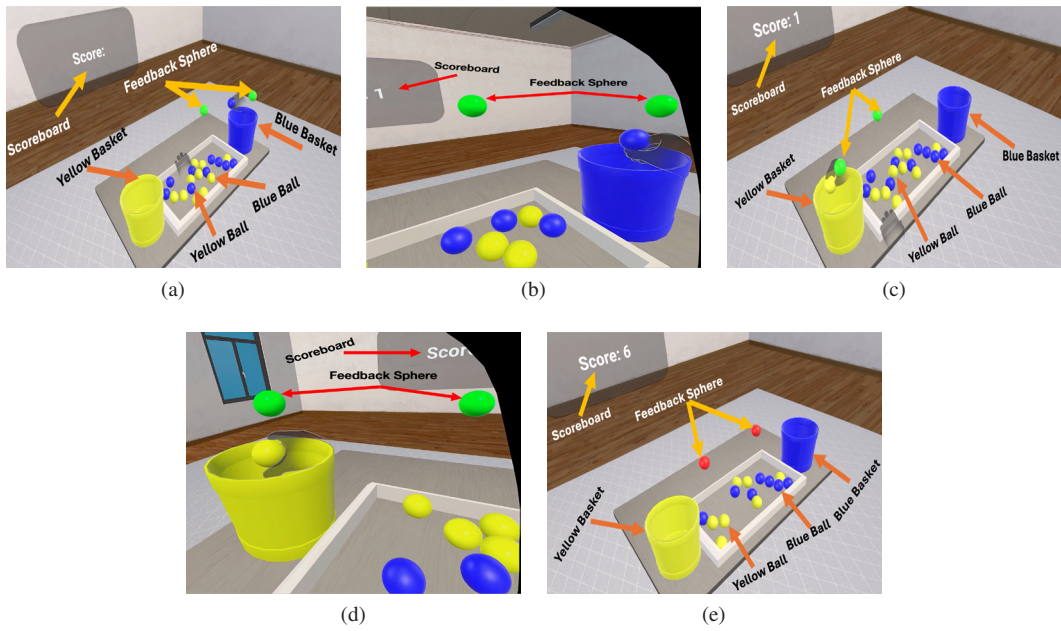


Fig. 5: a) A participant is placing a blue ball into the blue basket, and the hand tracking algorithm properly tracks both hands. The feedback spheres are green (highlighted by a light yellow arrow, with the basket and balls highlighted in orange). b) The image represents user perspective when they were placing the blue ball into blue basket (feedback spheres and scoreboard is highlighted in red arrow). c) The participant places a yellow ball into the yellow basket while the hand tracking algorithm continues to track both hands. The feedback spheres remain green (highlighted by a yellow arrow, with the basket and balls highlighted in orange). d) User perspective when they were placing the yellow ball into yellow basket (feedback spheres and scoreboard is highlighted in red arrow). When the hand tracking algorithm fails to track the hands, the feedback spheres turn red for both hands, and the system is unable to render any virtual hands, as their position and rotation cannot be determined. The black section on the corner of images b) and d) is rendered by the software when the user view is displayed on a 2D computer monitor. Participants did not see these black sections during the experiment.

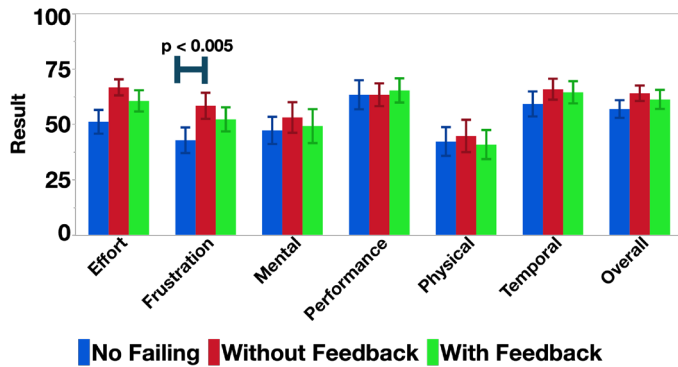


Fig. 6: The NASA TLX results for the Cup Stacking task were analyzed across the three tracking failure conditions: No Failing, With Feedback, and Without Feedback.

4.2.2 Procedure

We designed a virtual environment that included a table, along with B_2 : blue and yellow colored balls, and their corresponding baskets for the sorting task. The objective was for participants to pick up the balls and place them in the matching colored baskets. The study featured three different tracking failure conditions: TF_3 : no failing, with feedback, and without feedback as before.

The overall procedure (briefing, consent, demographics, headset setup, and familiarization) followed the same protocol as in Study 1, with participants seated facing a blank wall and wearing a Meta Quest Pro VR headset Sec. 4.1.3. The entire experiment took under an hour.

4.2.3 Virtual Environment

In each tracking failure condition, participants were positioned in front of a virtual wooden table, which featured a rectangular area filled with ten blue and ten yellow balls. Corresponding colored baskets were placed on either side of this area. A countdown timer was displayed, and participants were instructed to start the task when the timer reached “GO.” A scoreboard display was shown 133 cm above and 116 cm away from the table to track the number of correctly sorted balls, with the count only increasing when a ball was successfully placed in its corresponding basket, at which point a sound was played. If a ball was dropped or fell off the table, it was replaced with a new one, accompanied by an alert sound. When selecting a ball, the virtual hand would grasp it as naturally as a real hand would. Notably, no haptic feedback was provided during interactions with the virtual objects, meaning participants relied solely on their bare hands for all interactions. Each participant completed the task once in each condition.

The virtual spheres, configured identically to those in Study 1, were placed to provide an early warning based on the tracking status for each hand Fig. 5. These feedback spheres were attached to the participant’s virtual head (i.e., aligned with the headset), ensuring that they were visually and spatially distinct from the sorting balls on the table and contextually separate from the main pick-and-place activity. Consequently, participants could easily differentiate the feedback spheres from the task-related objects, reducing the risk of confusion.

4.2.4 Hand Tracking Failures

In the experiment, hand tracking failures were deliberately and randomly introduced *four times* during both the “with feedback” and “without feedback” conditions. Thus, we did not record an additional error variable. The randomness of these failures was controlled based on the number of balls left to be sorted. The randomness of these failures was tied to the number of balls left to be sorted. Participants were required to sort a total of 20 balls, which were divided into *four*

zones—each consisting of *five balls*. The system determined, at random, when to signal the experimenter to introduce tracking failures within each zone. Once the task began, the system prompted the experimenter with a “lights off” alert, which was not visible to the participant. The experimenter immediately turned off the lights in response, as the switch was within easy reach. After a 2-second interval—based on pilot testing—the system issued a “lights on” alert, and the experimenter restored the lighting. This off-on cycle took approximately *two seconds* and was repeated four times until the participant finished sorting the 20 balls. This approach allowed us to maintain a controlled environment while incorporating an element of randomness in the timing of the tracking failures.

As in Study 1, we used the headset’s built-in function to monitor the hand tracking algorithm’s performance. We did not use any additional algorithms to vary the hand tracking quality of the headset. Moreover, we did not use any hardware, such as gloves, for tracking user hand movements.

The tracking failure condition (TF_3) was counterbalanced using a Latin Square design across 18 participants. After completing each condition, participants were asked to fill out the System Usability Scale (SUS) [39] and NASA Task Load Index (TLX) [32] questionnaires. The SUS consisted of 10 questions, which are provided in the supplementary materials.

4.2.5 Results

As in Study 1, we analyzed the quantitative data using ANOVA with post-hoc tests and subjective data using the Friedman and Wilcoxon Signed-Rank tests for post-hoc analyses. The results are presented in Tab. 3, and Fig. 8 - Fig. 9. For brevity, we focus on the statistically significant results.

Table 3: Time taken to complete the ball sorting task under each tracking failure condition.

Parameters	No Failing	With Feedback	Without Feedback
Time	32.98 ± 7.86	41.35 ± 7.5	48.37 ± 9.07

Time: According to the results shown in Tab. 3 and Fig. 7, we observe a significant difference in time across the three conditions NO FAILING, WITH FEEDBACK, WITHOUT FEEDBACK, ($F(2,34) = 81.741, p < 0.001, \eta^2 = .828$). Specifically, participants took the least amount of time in the NO FAILING condition ($M = 32.98, SD = 7.86$), and the most time in the WITHOUT FEEDBACK condition ($M = 48.38, SD = 9.08$). Post hoc comparisons using the Bonferroni correction revealed that all pairwise comparisons between conditions were significant ($p < 0.001$).

These results suggest that the presence of visual feedback significantly affects task performance, with the WITHOUT FEEDBACK condition leading to the slowest completion times.

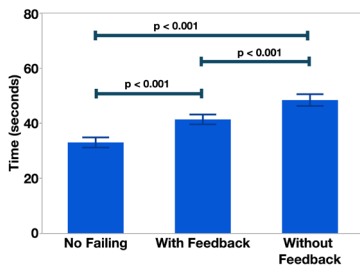


Fig. 7: Ball Sorting task completion time (in seconds) for tracking failures condition (No Failing, With feedback, Without feedback).

SUS: Based on the SUS scores presented in Tab. 4, there was a decrease in the SUS score for the WITHOUT FEEDBACK tracking failures condition, similar to what was observed in Study 1. Specifically, we observed an average SUS score of 60.41 ($Grade = D$) for WITHOUT FEEDBACK, which falls under the ‘Poor’ rating category.

Table 4: SUS Scores and Grades for three tracking failures condition for Ball Sorting task.

Condition	Score	Grade
No Failing	92.08	A
With feedback	89.86	A
Without feedback	60.41	D

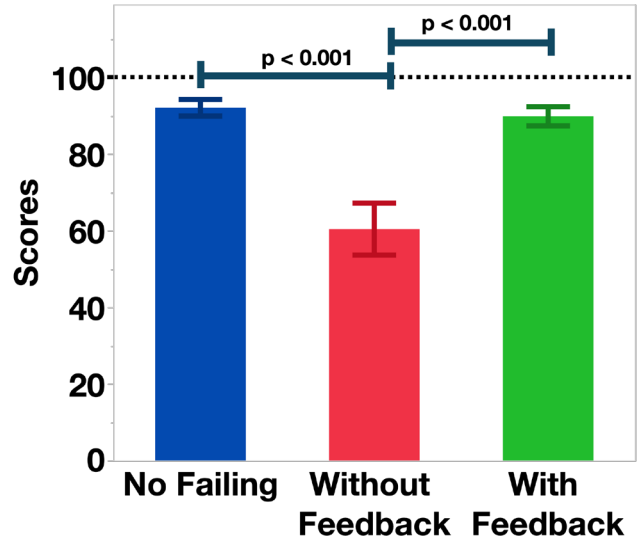


Fig. 8: SUS questionnaire results for Ball Sorting task for the three tracking failure conditions: No Failing, With Feedback, and Without Feedback.

We analyzed the overall SUS scores using a Friedman Test, revealing a significant difference across the three conditions, NO FAILING, WITHOUT FEEDBACK, and WITH FEEDBACK ($\chi^2(2,18) = 31.508, p < 0.001$). Post-hoc Wilcoxon Signed-Rank Tests revealed significant difference between NO FAILING and WITHOUT FEEDBACK ($Z = -3.726, p < 0.001, \eta = 0.88$), and between WITHOUT FEEDBACK and WITH FEEDBACK ($Z = -3.624, p < 0.001, \eta = 0.85$). However, no significant difference was found between NO FAILING and WITH FEEDBACK ($Z = -1.622, p = 0.105, \eta = 0.38$), as shown in Fig. 8. These findings support our hypothesis, **H1**, that an early warning system enhances VR usability when hand tracking failures occur during bare hands interaction.

NASA TLX: We used The Friedman Test to compare NASA TLX scores across the three conditions: NO FAILING, WITHOUT FEEDBACK, and WITH FEEDBACK—revealing a significant overall difference ($\chi^2(20,18) = 52.256, p < 0.001$). Further analysis of individual workload subscales indicated a significant difference in temporal demand ($\chi^2(2,18) = 9.043, p < 0.011$). The Post-hoc analysis via Wilcoxon Signed-Rank Test revealed a significant difference between the NO FAILING and WITH FEEDBACK conditions ($Z = -2.105, p < 0.035$), between WITHOUT FEEDBACK and NO FAILING ($Z = -2.753, p < 0.006$) conditions and between WITH FEEDBACK and WITHOUT FEEDBACK conditions ($Z = -2.202, p < 0.028$), as shown in Fig. 9. We also found a significant difference in frustration ($\chi^2(2,18) = 8.542, p < 0.014$). Post hoc analysis revealed significant differences between the WITHOUT FEEDBACK and NO FAILING conditions ($Z = -2.164, p < 0.03$) and between the WITH FEEDBACK and WITHOUT FEEDBACK ($Z = -2.969, p < 0.003$) conditions, see Fig. 9.

Participants took longer to complete the task and reported higher frustration in the WITHOUT FEEDBACK condition compared to the NO FAILING and WITH FEEDBACK conditions. This suggests that the lack of visual feedback on hand tracking failures increased task difficulty and led to increased frustration among the participants. These findings support our hypothesis **H2**, which states that an early warning system reduces users’ perceived task workload during bare-hand interactions

in a virtual environment.

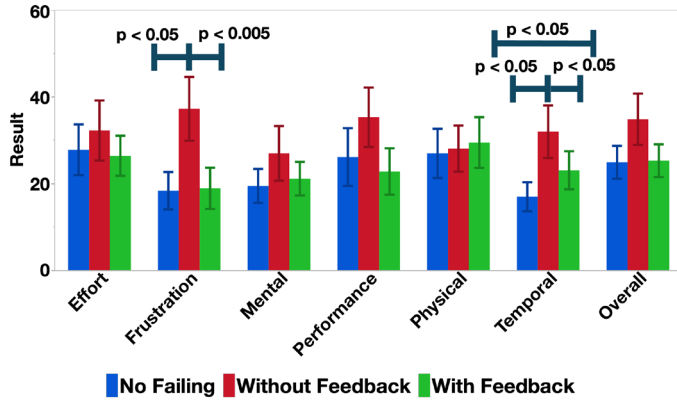


Fig. 9: Ball Sorting task's NASA TLX results for three tracking failures conditions: No Failing, With Feedback, and Without Feedback.

5 DISCUSSION

In this paper, we examined how visual feedback affects user perception and evaluated an early warning mechanism designed to address hand-tracking failures in VR head-mounted displays (HMDs). We conducted two user studies—one requiring participants to stack cups and another to sort colored balls—aiming to determine whether alerting users to potential hand-tracking failures in advance enhances performance, usability, and perceived workload during bare-hand interactions. Unlike recent studies that focus on post-error correction to improve hand-tracking reliability for fine motor tasks [1, 34], our work investigates the benefits of providing feedback *before* tracking failures occur. Even though we had 18 participants in each study, all the effect sizes were larger than $\eta^2 > 0.14$, indicating large effects. More importantly, our number of participants is consistent with previous work [9, 46], indicating robust findings and suggesting that our results are likely replicable.

In Study 1, we found that completion time did not significantly differ across the three conditions—(NO FAILING, WITH FEEDBACK, WITHOUT FEEDBACK)—for both stacking types (PILLAR SHAPE, TRIANGLE SHAPE). This suggests that the presence or absence of feedback did not affect the overall task completion time. The SUS scores revealed a significant difference in overall usability across the three conditions. Participants in the WITHOUT FEEDBACK condition reported a greater need for technical support, perceived more inconsistencies, and received a notably low average SUS score (*Grade D*). This indicates that, despite comparable task completion times, users found the system significantly less usable when no feedback was provided—highlighting the importance of feedback in enhancing user confidence and perceived system reliability.

The time results of Study 2 showed differences with those from Study 1. Participants took significantly longer to complete the task in the WITHOUT FEEDBACK tracking failures condition compared to both the NO FAILING and WITH FEEDBACK tracking failures conditions, demonstrating that visual feedback improves task efficiency. A similar pattern emerged in the SUS results, which revealed a significant difference in overall usability scores among the three conditions, similar to the findings in Study 1. They also perceived the system as more inconsistent in the WITHOUT FEEDBACK tracking failure condition and reported feeling less confident using the system compared to the NO FAILING tracking failures condition.

The findings on time results and SUS scores highlighted the importance of early warning systems in improving user performance and system usability. Based on this, we demonstrated that providing an early warning about impending hand-tracking failures improved participants' SUS scores and boosted their confidence in the system. This outcome supports our hypothesis **H1**: *an early warning system with visual feedback improves the usability of the virtual reality system.*

Beyond performance gains, our NASA TLX data from both studies reveal significantly reduced frustration and perceived time pressure when warnings are present in WITH FEEDBACK tracking failures condition. This suggests that participants experience less cognitive load because they can anticipate and plan their next moves rather than respond reflexively to a sudden failure. In other words, providing advanced notice enables them to allocate attentional resources more effectively. They do not have to continuously monitor for an unseen error; instead, they can rely on the system to notify them if a hand-tracking failure is imminent. This proactive approach can be explained by cognitive load theory [64], which states that reducing unexpected demands or “extraneous load” preserves mental resources, enabling more attention for the primary task [19]. Consequently, participants feel more confident and experience fewer disruptive moments that require sudden mental shifts or strategy changes. By giving users time to anticipate a potential loss in hand tracking—rather than forcing them to react to a sudden failure—the system lowered perceived task workload, thereby supporting our hypothesis **H2**, which states that *an early warning system reduces the user's perceived task workload while interacting using bare hands within a virtual environment.*

Although both tasks in this study are pick-and-place, each introduces unique dynamics, difficulty levels, and cognitive demands. That's why, e.g., study 1 and study 2 time results were different. In both studies, participants reported higher frustration and provided lower usability scores when no feedback was given—highlighting the robustness of our system and its potential for wider applicability. These pick-and-place activities (sorting balls, stacking cups) closely resemble real-world setups that require basic hand-eye coordination and object manipulation [21]. Eye-hand coordination tasks of this kind are commonly found in various fields such as maintenance and construction [10, 33]. For instance, in maintenance, technicians often need to handle and organize tools and parts efficiently, a process that requires precise coordination and dexterity. Similarly, in construction, workers frequently engage in activities like sorting and stacking materials or assembling components, which demand similar skills and attention to detail. The early warning visual feedback system can be employed to enhance the safety and efficiency of both fields, reducing frustration by presenting crucial information directly in the users' field of view and enabling them to focus on the task at hand without being overwhelmed by complex instructions. Consequently, adopting this approach in maintenance and construction contexts may mitigate errors, speed up workflow, and enhance user confidence, highlighting the proposed solution's practical impact and broader relevance.

In this paper, we investigated hand tracking failures that occur when there is no light in the environment, such as during power outages, which are a common part of daily life in many settings. Previous research has shown that hand tracking performance significantly decreases in environments with insufficient light sources [25, 51]. For example, Erol et al. highlighted the detrimental effects of uncontrolled environments, such as variable lighting and on-hand pose estimation [24]. Given that low-light conditions are a common and critical issue affecting hand tracking, our results can be used to improve user experience in different scenarios.

Our study highlights the importance of feedback mechanisms in improving users' perceptions of the system and reducing frustration. In particular, we provided a simple early warning visual cue whenever hand-tracking failure risk was imminent—specifically as the lights were dimmed—which allowed participants to prepare or adjust their actions to avoid failure. We deliberately chose this minimal design not to disrupt users mid-task. The warning supported their sense of agency by aligning expectations with reality and letting users reframe the failure as manageable rather than catching them off guard with an unexpected loss of input fidelity; it follows the same principle as a low-battery warning on a computer, which merely alerts the user but does not plug in the cable for them. In the presence of feedback, participants often paused or took corrective actions upon seeing the warning; in its absence, participants remained unaware, made mistakes, and eventually became more frustrated. Consequently, those without feedback took longer to finish. These observations led to lower frustra-

tion levels and an improved perception of the system in the feedback condition. We speculate that during the early warning phase, participants anticipated potential tracking loss, planned their next steps, and stayed more engaged, ultimately feeling more competent in completing the task. Because they did not have to rapidly switch strategies at the last second, their cognitive load likely decreased, enabling better task execution. As a result, system usability increased, highlighting how early warning systems can be essential for maintaining efficiency in VR scenarios that demand high precision and accuracy.

Our results align with earlier work showing that visual feedback can enhance user experience and perceived control in virtual environments [18, 56]. However, while previous studies predominantly examined feedback during the act of grasping, our approach focuses on preemptive feedback for impending tracking failures. By warning users before hand tracking is lost, we reduce performance disruptions and strengthen user confidence, extending the benefits highlighted by earlier research.

Moreover, the positive impact on user confidence and task consistency suggests that the proposed method could be beneficial in a wide range of VR applications beyond gaming and entertainment. For instance, medical training simulations that require precise hand movements or virtual prototyping and design tasks where accuracy and user satisfaction are vital may benefit from integrating an early warning visual feedback system. By alerting users to changes in hand-tracking fidelity—without overburdening them with complex instructions—developers can enhance VR applications' reliability, improve user experience, and encourage more efficient workflows.

6 DESIGN RECOMMENDATIONS

To enhance user interaction in VR systems, we recommend implementing early warning visual feedback mechanisms to alert users of hand tracking failures, ensuring that stimuli are placed within the user's field of view. This improves user perception of the system and reduces frustration, as shown by user experience results. The feedback system should be robust against varied lighting conditions and explore additional modalities like auditory or haptic feedback for a multi-sensory experience. Several visual stimulus methods, such as changing the color of hands, changing the color of the task, or adding text in distal space, can be potential implementation methods.

7 LIMITATIONS AND FUTURE WORK

This study used a specific VR HMD (Meta Quest Pro), and its hand tracking may not generalize to other devices. We recommend evaluating the early warning method on different HMDs to confirm its broader applicability. Without access to the internal hand-tracking algorithm, we relied on the manufacturer's data. Although we tested devices like the Meta Quest 2 and 3, none provided detailed confidence metrics. Future research could leverage built-in device tracking variables (if available) and include more diverse demographics, as factors like skin tone may influence the experience of early warning systems.

While advanced, hand tracking remains imperfect under rapid or occluded movements [3, 62], which we did not assess. We focused on one form of visual feedback; exploring other modalities (e.g., auditory, haptic) or varying feedback placement could lead to more comprehensive solutions. Our primary aim was to show that an early warning system improves usability and reduces task load; follow-up studies could investigate different setups in VR and AR, along with tasks involving greater cognitive demands (e.g., complex assembly or interactive learning).

We evaluated our method through two user studies focusing on eye-hand coordination. While the second user study included additional decision-making steps—such as selecting which balls go into which basket—it would be beneficial to assess the system's effectiveness with even more complex tasks beyond pick-and-place actions. Tasks with heavier cognitive loads may yield different results and further highlight the advantages of the proposed method. Testing this feedback system in contexts like complex assembly tasks provides valuable insights into its strengths and limitations.

We selected red and green as feedback colors (similar to traffic signals) and did not account for users with color vision deficiencies.

Testing different color schemes and examining intermittent or event-based notifications could further refine this method. Additionally, the interval between the warning and the tracking failure was not guaranteed to be identical for every occurrence due to human error. Finally, we did not instruct users on handling hand-tracking failures; instead, we opted to see how they naturally noticed and responded to the warnings. Providing explicit guidance in future work may further boost user confidence and overall system usability.

Finally, we see potential in incorporating built-in hand tracking machine learning algorithms to predict and compensate for tracking failures before they disrupt user experience. As newer headsets and algorithms emerge, re-testing and extending our approach could advance the usability of VR systems.

8 CONCLUSION

In this paper, we investigated the effects of integrating a visual feedback system as an early warning mechanism to address hand tracking recognition failures in VR HMDs. We conducted two user studies to evaluate the effectiveness of this approach. In the first study, participants stacked cups to form a pillar and a triangular shape, while in the second study, they sorted colored balls into corresponding baskets. Through these studies, we evaluated the effectiveness of a visual early warning system in informing users about tracking failures during interactions via hand within a virtual environment. The findings from our research suggest that providing early warnings before the hand tracking system fails improves the overall usability of the VR system. We hope that our proposed method can be used as an assistive system by alerting users to potential tracking errors; such systems can allow for immediate corrective action, thereby enhancing interaction quality and system reliability.

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